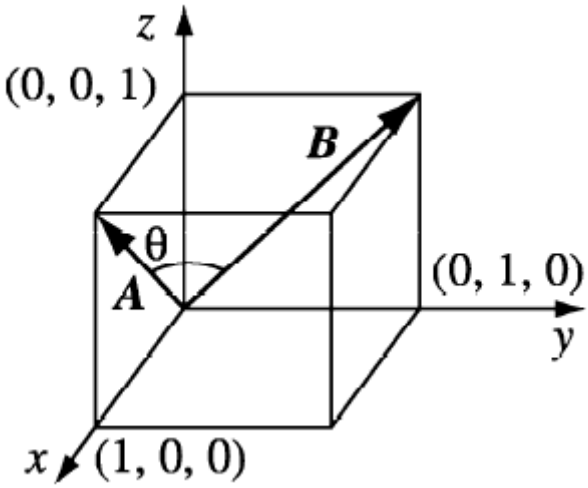
Example 1.2: Angle Between Face Diagonals of a Cube

The problem is to find the angle between the face diagonals of a unit cube.



Coordinate System and Setup

We consider a unit cube with its edges aligned along the coordinate axes:

- The vertices have coordinates:

$$(0,0,0), (1,0,0), (0,1,0), (0,0,1), \ldots$$

- The face diagonals are chosen as follows:
  1. One diagonal in the yz plane: connecting $(0,0,0)$ and $(0,1,1)$.
  2. Another diagonal in the xz plane: connecting $(0,0,0)$ and $(1,0,1)$.

Vectors Representing the Diagonals

Define the vectors corresponding to the chosen diagonals:

$$\vec{A} = \hat{x} + \hat{z}, \quad \vec{B} = \hat{y} + \hat{z}.$$

Dot Product and Cosine of the Angle

The angle θ between the vectors is given by:

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}| \cos\theta.$$

Compute the dot product:

$$\vec{A} \cdot \vec{B} = (\hat{x} + \hat{z}) \cdot (\hat{y} + \hat{z}).$$

Expanding this:

$$\vec{A} \cdot \vec{B} = (\hat{x} \cdot \hat{y}) + (\hat{x} \cdot \hat{z}) + (\hat{z} \cdot \hat{y}) + (\hat{z} \cdot \hat{z}).$$

Using orthogonality ($\hat{x} \cdot \hat{y} = \hat{x} \cdot \hat{z} = \hat{z} \cdot \hat{y} = 0$) and $\hat{z} \cdot \hat{z} = 1$:

$$\vec{A} \cdot \vec{B} = 1.$$

Compute the magnitudes of $\vec{A}$ and $\vec{B}$:

$$|\vec{A}| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}, \quad |\vec{B}| = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}.$$

Substitute into the equation:

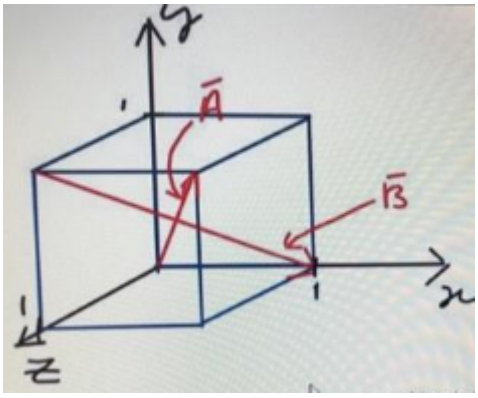
$$\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}.$$

Thus:

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ.$$

Problem 1.3: Angle Between Body Diagonals of a Cube

Extend the analysis to the body diagonals of the cube. Body diagonals are those connecting opposite corners, such as $(0, 0, 0)$ to $(1, 1, 1)$. Follow a similar procedure to calculate the angle between two



body diagonals:

1. Determine the vectors representing the diagonals.
2. Compute their dot product.
3. Use the magnitudes and the dot product to find $\cos\theta$.

Section 1.1.3: Triple Products

Scalar Triple Product

The scalar triple product of three vectors $\vec{A}, \vec{B}, \vec{C}$ is defined as:

$$\vec{A} \cdot (\vec{B} \times \vec{C}),$$

where:

- $\vec{B} \times \vec{C}$ produces a vector perpendicular to the plane of $\vec{B}$ and $\vec{C}$.
- $\vec{A} \cdot (\vec{B} \times \vec{C})$ gives the volume of the parallelepiped formed by $\vec{A}, \vec{B}, \vec{C}$.

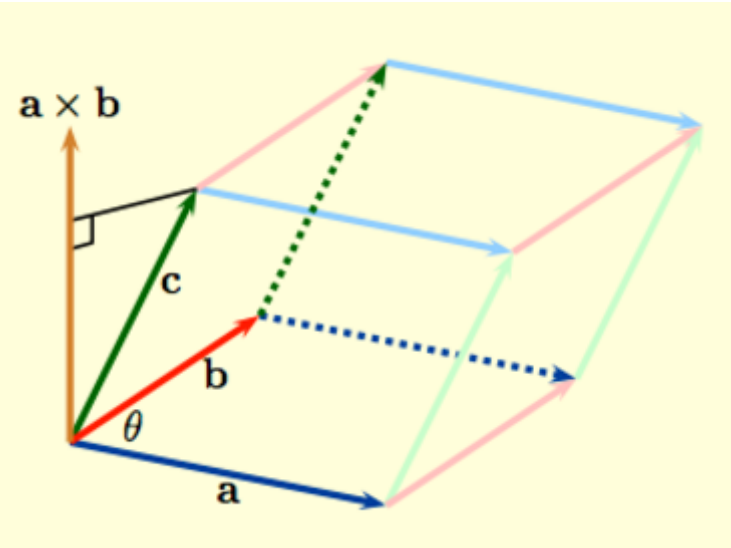
Vector Triple Product

The vector triple product is given by:

$$\vec{A} \times (\vec{B} \times \vec{C}),$$

which expands using the **BAC-CAB** rule:

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}.$$

Illustration with Parallelogram

A parallelogram is used to represent the scalar and vector triple products geometrically. The direction of $\vec{B} \times \vec{C}$ is normal to the parallelogram spanned by $\vec{B}$ and $\vec{C}$.